Vol 3 Chapter 10 — Atomic Clocks: Koons Tick + Substrate-Coordinate Precision

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Chapter 10 — Atomic Clocks

Atomic clocks are the most precise timekeeping instruments humanity has built. Optical lattice clocks based on Sr-87 and Yb-171 reach fractional precision near 10^{-18} — equivalent to losing one second over the age of the universe. The next-generation nuclear-transition clocks (Th-229) target 10^{-19}.

BST predicts atomic-clock measurements via the substrate's natural temporal granularity — the Koons tick $t_{\text{substrate}} = t_{\text{Planck}} \cdot \alpha^{C_2^2} \approx 10^{-120}$ seconds (Volume 0 Chapter 3 §3.2).

10.1 Substrate-coordinate precision

Atomic-clock measurements probe the substrate's coordinate structure at scales above the Koons tick but well below standard atomic transition times. The substrate's natural per-tick state $GF(128)^k$ Reed–Solomon codespace (Volume 0 Chapter 3 §3.5) sets a structural precision limit.

The framework's prediction: clock precision improvements beyond $\sim 10^{-19}$ will encounter substrate-coordinate corrections at the parts-per-billion-of-precision level. This is at the edge of current technology; multi-year experimental targets.

10.2 SP-30-4 time-granularity falsifier

The SP-30-4 substrate engineering proposal (Lyra theoretical first, May 2026, 200-400K experimental program) targets direct measurement of the substrate's time-granularity correction at 10^{-19} precision via next-generation optical lattice clocks. The prediction is that the substrate's $N_c \cdot t_{\text{Planck}}$ clock-cycle structure produces measurable deviations from standard QED at this precision.

The substrate-mechanism reading: the substrate's discrete tick imposes a fundamental precision floor below which atomic-clock measurements encounter substrate-coordinate quantization.

10.3 What comes next

Chapter 11 — Nuclear Decay — develops the substrate's g-field-exponent structure in the weak nuclear-decay sector.

Where to look this up: T2405 (Koons tick). SP-30-4 (time-granularity falsifier). For standard atomic-clock physics: Ludlow et al. 2015 review *Reviews of Modern Physics* 87:637.